

Chapter 4: Electricity and Magnetism

Section 4.1: Electric Charges and Fields

Electric charge is the fundamental property of matter that causes it to experience a force when placed in an electromagnetic field. Charges are of two types: positive and negative. Like charges repel and opposite charges attract. The SI unit of electric charge is Coulomb (C). Coulomb's Law quantifies the electrostatic force between two point charges.

Section 4.2: Electric Current and Flow of Electrons

Electric current (I) is defined as the net rate of flow of electric charges across a given cross-sectional area of a conductor. Mathematically, $I = Q / t$, where Q is the net charge flowing in time t . The SI unit of electric current is the Ampere (A), named after André-Marie Ampère. One ampere corresponds to the flow of one coulomb of charge per second.

Section 4.3: Electric Potential and Potential Difference

Electric potential difference (V) between two points in an electrical circuit is the amount of work done (W) in moving a unit positive charge from one point to the other. $V = W / Q$. The SI unit of potential difference is Volt (V). A battery or power source provides the required potential difference to drive steady electric current through an electrical circuit.

Section 4.4: Ohm's Law and Electrical Resistance

Georg Simon Ohm discovered in 1827 that the current flowing through a metallic conductor is directly proportional to the potential difference across its terminals, provided the temperature remains constant. Mathematically, $V = I * R$. The constant of proportionality R is called the Resistance of the conductor. Its SI unit is the Ohm (?).

Section 4.5: Factors Affecting Electrical Resistance

The electrical resistance of a uniform metallic conductor depends directly on its length (L), inversely on its area of cross-section (A), and on the resistivity (ρ) of the material: $R = \rho * (L / A)$. Resistivity is a characteristic property of the material and varies with temperature. Conductors have low resistivity, while insulators have high resistivity.

Section 4.6: Resistors in Series Combination

When two or more resistors are connected end to end consecutively, they are said to be in series combination. In a series circuit, the same electric current passes through every resistor. The total or equivalent resistance (R_{eq}) is equal to the sum of individual resistances: $R_{eq} = R_1 + R_2 + R_3$. The total voltage equals the sum of voltage drops.

Section 4.7: Resistors in Parallel Combination

When resistors are connected across the same two common nodes, they are in parallel combination. The potential difference across each parallel resistor is identical. The reciprocal of the equivalent resistance is equal to the sum of the reciprocals of the individual resistances: $1 / R_{eq} = 1 / R_1 + 1 / R_2 + 1 / R_3$. The total current is divided among branches.

Section 4.8: Heating Effect of Electric Current (Joule's Law)

When an electric current flows through a purely resistive conductor, electrical energy is dissipated entirely as heat. According to Joule's Law of Heating, the heat generated (H) is proportional to the square of current, resistance, and time: $H = I^2 * R * t$. This principle is utilized in electric heaters, toasters, irons, and electrical safety fuses.

Section 4.9: Electrical Power and Energy Consumption

Electric power (P) is the rate at which electrical energy is consumed or converted into other forms of energy in an electric circuit. $P = V * I = I^2 * R = V^2 / R$. The SI unit of power is Watt (W). Commercial electrical energy is measured in kilowatt-hours (kWh), where $1 \text{ kWh} = 3.6 \times 10^6 \text{ Joules}$, commonly referred to as one unit of electricity.

Section 4.10: Chapter Summary and Practice Problems

Key Takeaways:

1. Electric current is charge per unit time: $I = Q / t$.
2. Ohm's Law relates voltage, current, and resistance: $V = I * R$.
3. Series resistance adds linearly, while parallel resistance decreases equivalent resistance.
4. Practice Problem 1: Calculate current through a 24 Ohm resistor connected to a 12V power supply.